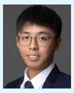
LandMark Optoelectronics

Outperform

Initiation: Ready to step into a supply gap

Semicons & Semicon Equip
Taiwan


Nick Hung


Arthur Lai

3081 TT	**Outperform**
Price (at 13 Nov 2025)	**TWD433.00**
12-month target	**TWD480.00**
12-month TSR (%)	**11.1**
Volatility Index	**Very High**
Market Cap (Local) (m)	40,062
Market Cap (USD) (m)	1,291
Free Float (%)	91

Key Points

- Amid booming transceiver demand and an EML shortage, the likely rise of SiPh transceivers bodes well for LandMark as a dominant CW laser supplier.
- InP substrate crisis and downstream yield issues have been resolved and its 3" laser qualification is on track with expansion plans ahead.
- We initiate coverage at Outperform on a NT$480 TP (35x 27E EPS).

- We forecast a 70% transceiver demand CAGR over 2024-27E, driven by AI scale-out networking. LandMark is a supplier of optical chips, such as laser diodes and photodetectors, used in transceivers. It possesses end-to-end optical chip manufacturing capabilities, from epiwafer to foundry to back-end process, enabling it to obtain direct business from one of the major US CSPs. We expect Datacom revenue will drive performance in 2026E, accounting for more than 85% of the firm's top line for the year and resulting in 153% EPS growth.
- We expect tighter supply of laser chips in 2026E, as the industry is unprepared for the non-cyclical fast ramp up in AI clusters and networking. We believe supply constraints of high-end EML lasers will lead to higher adoption of CW lasers, used in silicon photonics (SiPh) transceivers, as an alternative.
- LandMark will be a first-wave beneficiary of CW laser adoption as an upstream supplier with a proven track record, having already secured a US CSP customer.
- A recent slowdown in its Datacom revenue is due to 1) InP substrate ban from China leading to 2" to 3" epiwafer transition and 2) defects caused by its downstream partners. We expect both situations to be resolved by end of 1Q26E. We expect its revenue to pick up with 3" qualification and view the NT$360mn of capex announced for 2026 as indicative of management's confidence in its ability to deliver and its order visibility.

Valuation: Our NT$480 target price is based on 35x 2027E EPS, around its average since 2022. We highlight our 86% EPS CAGR over 2026-27E as additional support.

Catalysts: 3" laser epiwafers pass qualification, downstream yield improvement.

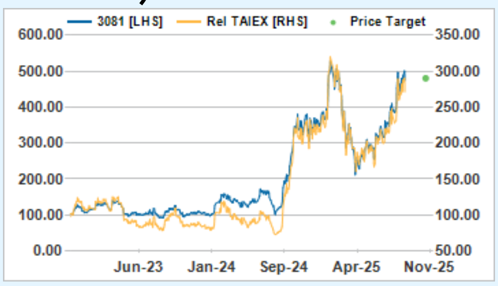
Investment Thesis and Recommendation

We expect LandMark to benefit from SiPh transceiver adoption as transceiver demand soars and the current laser solution experiences a supply crunch.

Investment Fundamentals

Year end 31 Dec	2024A	2025E	2026E	2027E
Revenue (m)	1,208	2,195	3,374	4,582
Revenue growth (%)	14.4	81.6	53.7	35.8
EBIT (m)	(111.8)	433.0	1,066	1,490
EBIT growth (%)	59.9	NM	146.1	39.8
Reported profit (m)	(54.5)	362.6	917.0	1,256
Adjusted profit (m)	(54.5)	362.6	917.0	1,256
EPS rep [TWD]	(0.6)	4.0	10.0	13.7
EPS rep growth (%)	74.3	NM	153.1	37.0
EPS adj [TWD]	(0.6)	4.0	10.0	13.7
EPS adj growth (%)	74.3	NM	153.0	37.0
Net debt/equity (%)	(14.0)	(8.2)	(13.1)	(27.0)
ROA (%)	(2.6)	9.5	20.2	23.6
ROE (%)	(1.4)	9.2	20.1	22.6
PER rep (x)	(727.8)	109.6	43.3	31.6
PER adj (x)	NM	109.5	43.3	31.6
EV/EBITDA (x)	153.3	57.3	30.9	23.4
P/BV (x)	10.5	9.6	8.0	6.5
Total div yield (%)	0.1	0.1	0.2	0.9

3081 TT rel TAIEX performance, & rec history



Source: FactSet, Macquarie Research, Nov 2025 (all figures in TWD unless noted, TP in TWD)

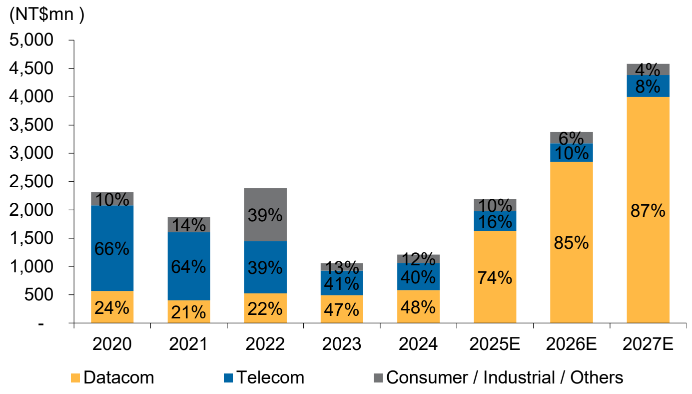
Key Risks to Investment Thesis

- Slower-than-expected transceiver demand.
- Over-capacity of leading players
- Failed or delayed 3" qualification

Company Description

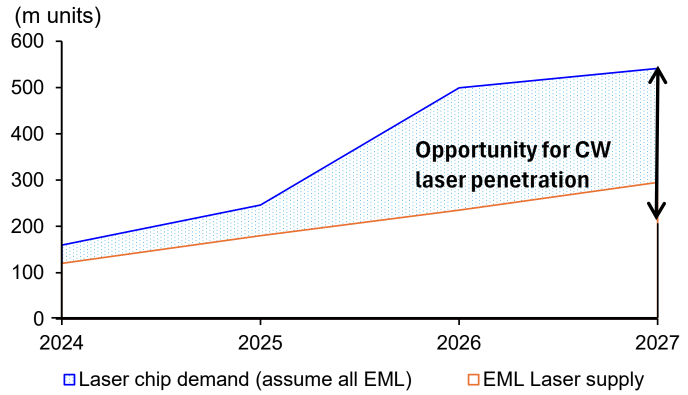
LandMark is a key supplier for CW lasers used in SiPh transceivers in data centres. The company mainly focuses on epitaxy, a raw material, but is also capable of delivering optical chips as turnkey products. The firm formerly worked with Intel on SiPh technology, and is able to secure orders directly from leading transceiver makers and US CSPs.

Figure 1 - LandMark revenue breakdown



Source: Company, Macquarie Research, Nov 2025

Figure 2 - EML laser shortage gives way to CW laser penetration



Source: Macquarie Research, Nov 2025

LandMark Optoelectronics (3081 TT) TWD/(m) unless otherwise noted

Income Statement Dec FY	2024A	2025E	2026E	2027E	Q3/25A	Q4/25E
Revenue	**1,208**	**2,195**	**3,374**	**4,582**	**552.8**	**634.1**
Cost of Goods Sold	969.5	1,291	1,701	2,267	311.6	358.6
Gross Profit	**238.9**	**903.4**	**1,673**	**2,315**	**241.2**	**275.5**
EBITDA	**279.6**	**747.6**	**1,388**	**1,831**	**180.0**	**213.2**
Depreciation	389.3	313.0	320.3	339.9	75.9	77.3
Amortisation	2.1	1.6	1.6	1.6	0.4	0.4
EBIT	**(111.8)**	**433.0**	**1,066**	**1,490**	**103.7**	**135.6**
Net Interest Income	20.7	29.1	38.2	38.2	9.5	9.5
Associates	0.0	0.0	0.0	0.0	0.0	0.0
Exceptionals	0.0	0.0	0.0	0.0	0.0	0.0
Forex Gains / Losses	24.4	(31.3)	0.0	0.0	0.0	0.0
Other Pre-Tax Income	(1.5)	22.4	42.3	42.3	10.6	10.6
Pre-Tax Profit	**(68.2)**	**453.3**	**1,146**	**1,570**	**123.8**	**155.7**
Tax Expense	13.6	(90.7)	(229.3)	(314.1)	(24.8)	(31.1)
Net Profit	**(54.5)**	**362.6**	**917.0**	**1,256**	**99.0**	**124.5**
Minority Interests	0.0	0.0	0.0	0.0	0.0	0.0
Reported Earnings	**(54.5)**	**362.6**	**917.0**	**1,256**	**99.0**	**124.5**
Adjusted Earnings	**(54.5)**	**362.6**	**917.0**	**1,256**	**99.0**	**124.5**
Basic Shares Outstanding	91.7	91.7	91.7	91.7	91.7	91.7
Diluted Shares Outstanding	91.7	91.7	91.7	91.7	91.7	91.7
EPS (rep) [TWD]	(0.6)	4.0	10.0	13.7	1.1	1.4
EPS (adj) [TWD]	(0.6)	4.0	10.0	13.7	1.1	1.4
Total DPS [TWD]	0.5	0.5	1.0	4.0	0.5	0.0

Ratio	2024A	2025E	2026E	2027E		
Revenue Growth (%)	14.4	81.6	53.7	35.8	-	-
EBITDA Growth (%)	56.8	167.4	85.6	32.0	-	-
EBIT Growth (%)	59.9	NM	146.1	39.8	-	-
EPS Growth (adj) (%)	74.3	NM	153.0	37.0	-	-
Gross Profit Margin (%)	19.8	41.2	49.6	50.5	-	-
EBITDA Margin (%)	23.1	34.1	41.1	40.0	-	-
EBIT Margin (%)	(9.3)	19.7	31.6	32.5	-	-
Net Profit Margin (%)	(4.5)	16.5	27.2	27.4	-	-
Payout Ratio (%)	NM	12.6	10.0	29.2	-	-
PE (rep) (x)	(727.8)	109.6	43.3	31.6	-	-
PE (adj) (x)	NM	109.5	43.3	31.6	-	-
EV/EBITDA (x)	153.3	57.3	30.9	23.4	-	-
EV/EBIT (x)	(383.5)	99.0	40.2	28.8	-	-
Price/Book (x)	10.5	9.6	8.0	6.5	-	-
Total Div Yield (%)	0.1	0.1	0.2	0.9	-	-

Balance Sheet	2024A	2025E	2026E	2027E
Cash	529.1	339.5	654.7	1,659
Receivables	207.0	454.6	740.0	807.4
Inventories	346.3	625.0	888.9	973.8
Investments	0.0	100.0	100.0	100.0
Fixed Assets	1,665	1,522	1,562	1,582
Intangibles	500.3	464.8	463.2	461.6
Other Assets	1,043	1,314	1,314	1,314
Total Assets	**4,290**	**4,821**	**5,723**	**6,898**
Payables	41.5	74.9	106.5	116.7
Short Term Debt	0.0	0.0	0.0	0.0
Long Term Debt	0.0	0.0	0.0	0.0
Other Liabilities	477.1	629.2	629.2	629.2
Total Liabilities	518.6	704.1	735.7	745.9
Shareholders' Funds	3,831	4,147	5,019	6,183
Minority Interests	0.0	0.0	0.0	0.0
Other	(59.5)	(30.9)	(30.9)	(30.9)
Total S/H Equity	**3,772**	**4,116**	**4,988**	**6,152**
Total Liab & S/H Funds	4,290	4,821	5,723	6,898
Net Debt / Equity (%)	(14.0)	(8.2)	(13.1)	(27.0)
Interest Cover (x)	NM	NM	NM	NM
ROE (%)	(1.4)	9.2	20.1	22.6
ROA (%)	(2.6)	9.5	20.2	23.6
ROIC (%)	(2.7)	10.7	22.6	27.5

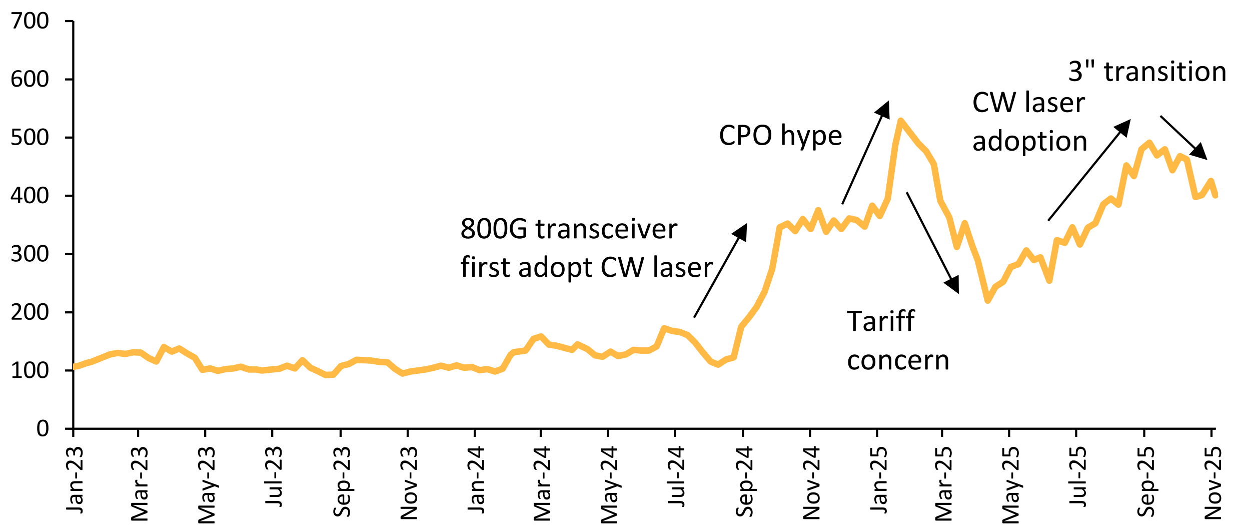
Cash Flow Analysis	2024A	2025E	2026E	2027E
EBITDA	**279.6**	**747.6**	**1,388**	**1,831**
Tax Paid	13.6	(90.7)	(229.3)	(314.1)
Chgs in Working Cap	27.0	(500.4)	(517.8)	(142.1)
Net Interest Paid	20.7	29.1	38.2	38.2
Other	35.7	166.6	42.3	42.3
Operating Cashflow	**376.5**	**352.2**	**721.1**	**1,456**
Acquisitions	(300.0)	(351.0)	0.0	0.0
Capex	(70.0)	(100.3)	(360.0)	(360.0)
Other	(11.9)	(40.5)	0.0	0.0
Investing Cashflow	**(382.0)**	**(491.8)**	**(360.0)**	**(360.0)**
Dividend (Ordinary)	(45.9)	(46.3)	(45.9)	(91.7)
Equity Raised	0.0	0.0	0.0	0.0
Debt Movements	0.0	0.0	0.0	0.0
Other	(1.7)	(5.1)	0.0	0.0
Financing Cashflow	**(47.6)**	**(51.4)**	**(45.9)**	**(91.7)**
Net Chg in Cash/Debt	**(51.5)**	**(189.7)**	**315.2**	**1,004**
Free Cashflow	**306.5**	**251.9**	**361.1**	**1,096**

Source: Company data, Macquarie Research Nov 2025

Investment thesis

- We are very positive on the transceiver market's prospects. As clusters scale to over 1mn + XPUs in 2027E, the required bandwidth for scale-out interconnect will grow rapidly with more layers of networking switches and transceivers needed; we forecast a 70% CAGR for the transceiver market over 2024-27E.
- We note signs of tightening supply in comments from downstream transceiver makers on shortages of electro-absorption modulated lasers (EML, used in high-end transceivers) as early as 2024. We expect the shortfall to widen in 2026E, which we believe will result in greater adoption of continuous-wave (CW) lasers in silicon photonic (SiPh) transceivers and new entrants in the upstream supply chain.
- We view LandMark as a first-wave beneficiary of CW laser adoption as an upstream supplier with proven track record with a US CSP customer. We expect LandMark to benefit from the massive adoption of SiPh transceivers, estimating its Datacom division revenue growth in 2026E at 75% YoY, driving up both the firm's overall revenue and its margins.
- We expect recent hiccups, including an indium phosphide (InP) ban, 3" transition and yield issues from downstream partners, will be resolved by the end of 1Q26E, setting up a positive 2026E as the firm rides rapidly expanding demand for transceivers amid a shortage of optical chips. We believe LandMark is also expanding its capacity to cater to the growing demand, given its announced NT$360mn capex on back-end equipment, suggesting stronger order visibility.

Figure 3 - LandMark stock price and key events



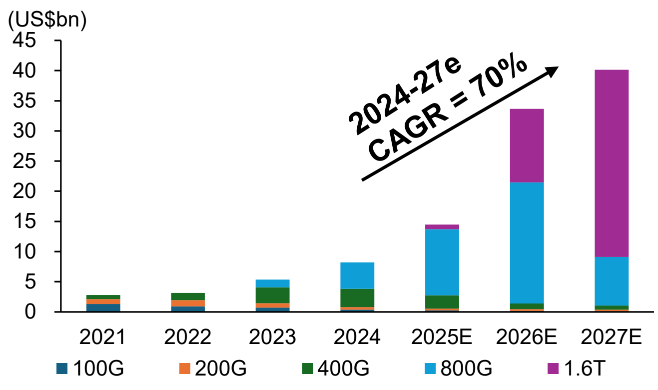
Source: TEJ, Macquarie Research, Nov 2025

Transceiver demand to surge starting 2H25

- We believe LandMark will be a major beneficiary of an explosion in demand for transceivers as data centres look to AI networking solutions to increase performance. We forecast a 70% transceiver market CAGR over 2024-27E, with growth skewed towards in 2026-27E.
- Transceivers require optical chips, including laser diodes (LD) and photodetectors (PD).
- LandMark manufactures epiwafers for optical chips, specifically for CW lasers used in SiPh transceivers in data centres. The company mainly focuses on epitaxy, which is a raw material, but it is also capable of delivering optical chips as turnkey products.
- We expect the company to record 75% Datacom segment revenue growth in 2026E, driven by the massive transceiver market growth and further adoption of CW laser solutions in SiPh transceivers.

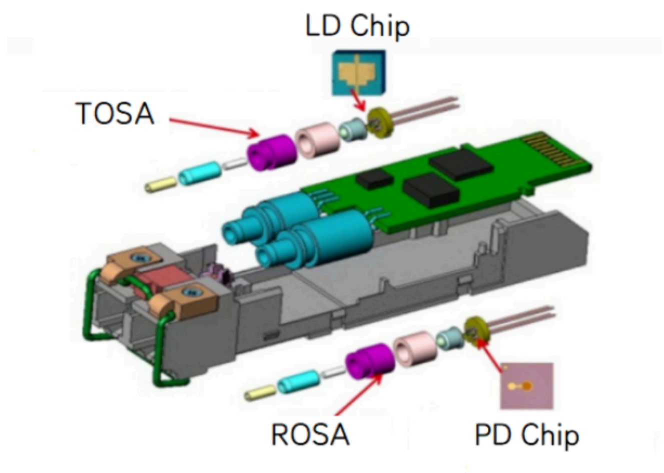
- LandMark has been a pioneer in CW lasers with comprehensive SiPh expertise, validated by its past supply history with Intel. It is qualified by a US CSP directly due to its capability of delivering full-stack solutions (Figure 7) and has also secured direct business with Innolight, the largest transceiver maker in the world.

Figure 4 - MQ transceiver market forecast



Source: IDC, Macquarie Research, Nov 2025

Figure 5 - Transceiver component overview



Source: Fibermall, Nov 2025

Figure 6 - Transceiver supply chain overview

Transceiver Component
- **Transceiver:** Innolight, Eoptolink, Fabrinet, Coherent, Lumentum, Accelink, AAOI, Ciena
- **TOSA/ROSA:** TFC, Elaser, Luxnet, Truelight, EZConn

Laser Diode (Laser chip) / **Modulator**
- **EML Laser:** Lumentum, Coherent, Sumitomo, Broadcom, Mitsubishi
- **CW Laser:** EML players, Sumitomo, Landmark, Yuanjie, Furukawa, AAOI
- **SiPh Modulator:** TSMC, UMC, Global Foundry, Tower semiconductor

Photodetector
- **Photodetector:** EML/CW players, GCS, Dexerials
- **Epi-wafer:** Landmark, VPEC

PCB
- **HDI:** SCC, Unimicron, ZDT, Compeq

Passive Components
- **Fiber /connector:** TFC, Senko, Fabrinet, FOCI, Browave, Sumitomo, Furukawa, Corning, Fujikura, EZconn

DSP
- **DSP:** Broadcom, Marvell, MaxLinear, Airoha

Source: Macquarie Research, Nov 2025

Well positioned in SiPh transceiver market

LandMark's experience in InP substrate, which is used in CW lasers, positions it as a first-wave beneficiary of rapid growth in transceiver demand. LandMark sources its InP substrate mainly from AXT China, and then uses MOCVD equipment to form epiwafers. The company offers two delivery models: epiwafers, its core business, and chip (turnkey) solutions for specific clients, if needed.

Q: What type of light source (laser) is needed in transceiver?

- Any optical communication requires light as a carrier for data. Silicon of itself is not a light source, so usually it needs a III-V compound semiconductor (InP or GaAs) to emit light. There are 3 types of lasers in high-speed transceivers, listed below:
  - ⇒ VCSEL: Uses a cheaper GaAs substrate. Almost impossible to manufacture for higher data rates. Direct modulation become too difficult. While low-speed VCSEL is easy to manufacture, high-speed VCSEL (used in 400G transceivers) is dominated by Broadcom.

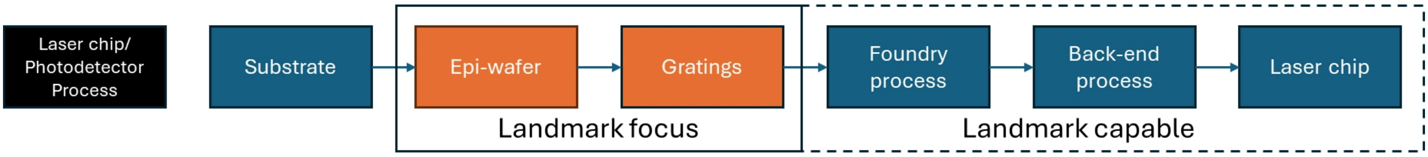
⇒ EML: the mainstream solution for high-speed, long-distance, high-quality optical communication, using expensive InP substrate with a difficult manufacturing process. It has always been the first choice, given its superior performance and proven reliability with an existing supply chain. Given the complexity of EML's structure, there are limited suppliers (see Figure 6).

⇒ CW laser: an alternative to EML, mainly used in SiPh transceivers. It is an EML without the modulation parts, which makes CW laser manufacturing much easier than EML laser production. CW laser modulation is done by SiPh chips.

Q: Why there is a laser industry shortage, and how do we expect it to play out?

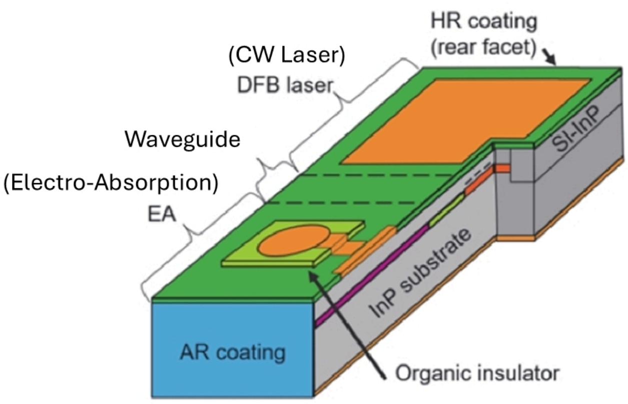
- We see evidence of a shortage in EML since 2024, from comments across the industry in earnings calls. The shortage has been driven primarily by surging demand for optical interconnect for high-speed networking.
- As we lay out in our ***AI Networking*** Ecosystem report, we believe the networking industry will exhibit exponential growth in 2026-27E, creating a non-cyclical demand uptick for components, resulting in a demand shock for the unprepared supply chain. The laser industry has been loss-making for a long time, even for leading players, and thus there has been a reluctance across the industry to expand capacity.
- We forecast demand for ~500mn EML chips in 2026E, assuming all transceivers use EML as the light source. However, our channel checks suggest only ~235mn EML chips can be produced. The EML laser shortage started in 2024E, but leading EML players' capacity expansion plans have been conservative, extending the shortage.
- Thus we expect the transceiver industry to shift away from EML and towards new SiPh solutions for their transceivers, as these only require CW lasers to function, allowing for more suppliers to compete.

Figure 7 - Optical chip manufacturing process - Landmark's role



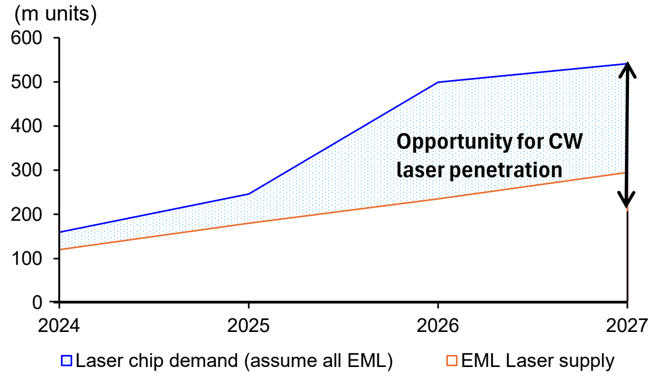
Source: Company, Macquarie Research, Nov 2025

Figure 8 - EML / CW laser structure



Source: Lumentum, Macquarie Research, Nov 2025

Figure 9 - EML laser shortage gives way to CW laser penetration



Source: Macquarie Research, Nov 2025

Figure 10 - Mainstream laser technology for optical communication

Feature	VCSEL (Vertical-Cavity Surface-Emitting Laser)	CW DFB (Continuous Wave Distributed Feedback)	EML (Electro-Absorption Modulated Laser)
Full Name	Vertical-Cavity Surface-Emitting Laser	Continuous Wave Distributed Feedback Laser	Electro-Absorption Modulated Laser
Basic Principle	Light resonates and is emitted **vertically** from the top surface of the chip.	Light resonates and is emitted from the **edge** of the chip. An internal grating filters for a single, stable wavelength.	An integrated chip combining two sections: a **CW DFB laser** (as the light source) and an **Electro-Absorption Modulator (EAM)**.
Modulation	**Directly Modulated (DML):** The laser is turned on and off by varying the drive current.	**Externally Modulated (Silicon Photonics):** This laser is modulated by an external Silicon Photonic chips. It is designed to output a stable, continuous stream of light. It is used *with* an external SiPh modulator.	**Externally Modulated (InP):** The DFB laser runs in CW mode (stable power). The separate EAM section acts as a "transcbriber", modulating the light to carry data.
Wavelengths	850 nm (primarily), 1310 nm	1310 nm, 1550 nm (O-band, C-band, L-band)	1310 nm, 1550 nm (O-band, C-band, L-band)
Fiber Type	**Multimode Fiber (MMF):** The 850nm light and beam shape are ideal for MMF.	**Single-Mode Fiber (SMF):** The narrow, stable wavelength is required for long-distance SMF.	**Single-Mode Fiber (SMF):** Required for the high-speed, long-distance performance this laser enables.
Data Rate	Up to 100 Gbps per lane	Up to 200 Gbps per lane (for 800G, 1.6T)	Up to 200 Gbps per lane (for 800G, 1.6T)
Transceiver usage	Widely used in 400G transceivers for Short-distance.	First wave of adoption in 400G, with higher adoption in 800G and 1.6T.	Widely used in high-speed transceivers across all ranges.
Transmission Distance	**Very Short Reach (VSR):** < 1 meter (AOCs) to ~100 meters.	Applicable for almost all distance.	**Long Reach (LR) / Extended Reach (ER, ZR):** 10 km to 80 km.

Source: Macquarie Research, Nov 2025

3" migration, more back-end equipment ready

We view LandMark's ability to resolve its recent hiccups by end of 1Q26E as positive, around which time we also expect the company to complete migration to 3", with 50%+ capacity upside. Additionally, we note preparations to add further capacity.

LandMark has seen demand increase since late-2024, as the EML shortage emerged. In 1H25 LandMark decided to introduce downstream partners after identifying its back-end process as a bottleneck to its delivery capability. However, LandMark's guidance for the past three quarters has been lagging that of laser chip industry leaders and downstream transceiver clients, with issues around defects associated with its outsourced products, delaying epiwafer shipments. The company has resorted to more suppliers and started to in-source some capacity, and we expect the issues will be resolved by 2026E.

A secondary reason for the firm's lagging performance was China's ban on InP substrate exports; LandMark imported most of its 2" epiwafer from China AXT, the only 2" supplier in the world. The ban was lifted in July, but, in order to avoid the risk of similar occurrences in future, the company decided to migrate to 3" InP epiwafer production, with the added benefit that 3" also offers higher output per wafer, bringing more capacity on-line. While it takes 6,000+ hours of laser burn-in testing of new capacity, we expect LandMark's 3" migration to finish in 1Q26E, with at least 50%+ Datacom revenue run rate upside. The newly announced NT$360mn of capex indicates management's confidence in delivering revenue as well as gaining more orders in the 2H26E. We are forecasting 75% Datacom revenue growth for 2026E, roughly in line with our expectation of transceiver industry annual compound growth at 70%.

Company overview

LandMark is a professional epitaxy manufacturer, with InP substrates its primary raw material and using MOCVD technology.

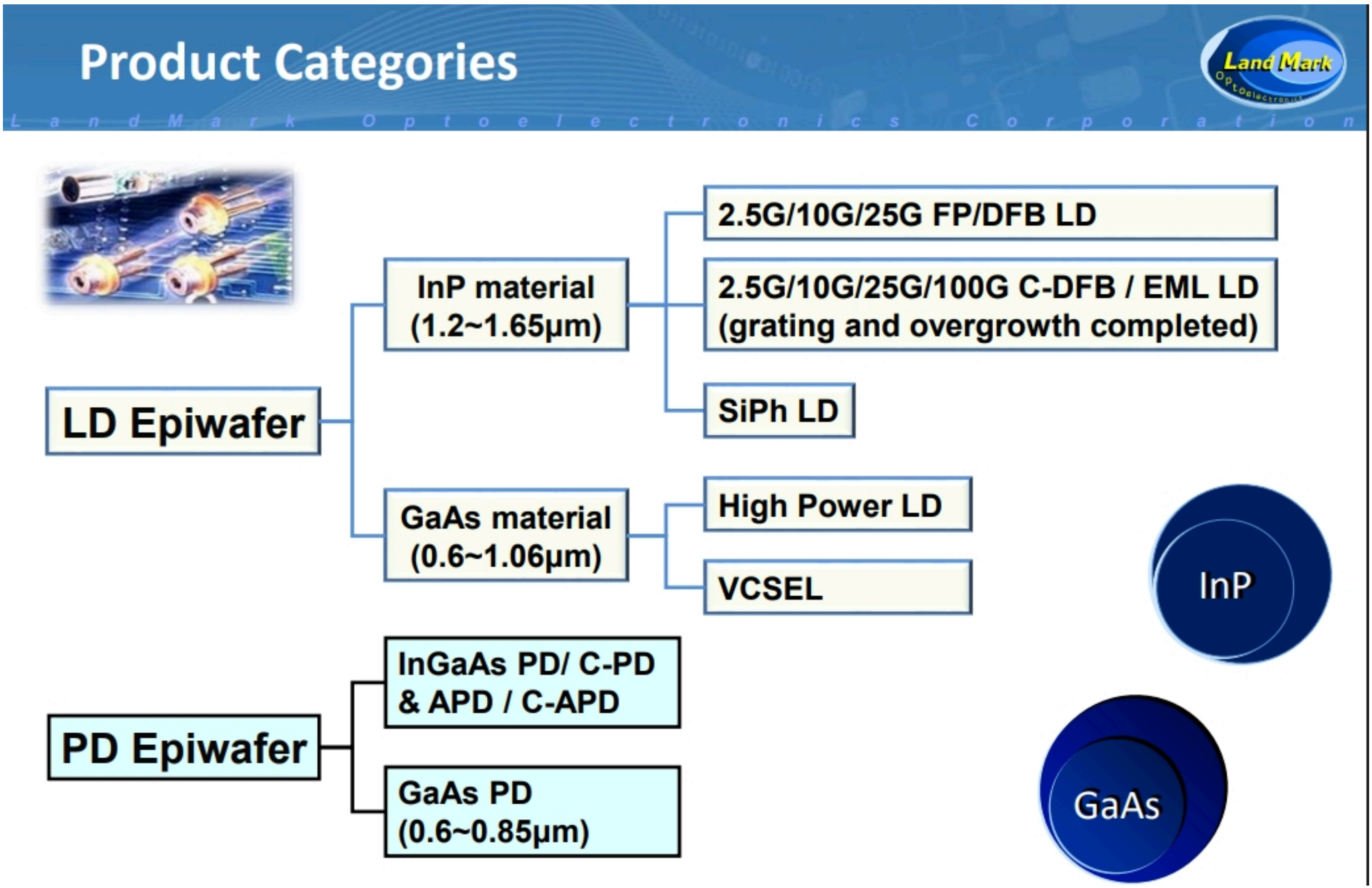
LandMark's business is divided into 3 segments: Datacom, Telecom, and 'Other'.

For Datacom, LandMark has extensive experience and orders; we expect the unit to be the company's major growth driver in 2025-27E, as we expect Datacom revenue to reach 87% of total revenue in 2027E. Other than InP CW lasers for high-speed transceivers, the company also ships photodetector (PD) dies pairing with its lasers, providing an end-to-end solution for its customers.

For its Telecom division, LandMark mainly ships 10G/25G DFB laser epiwafers, which are used for transmitters in GPON and 10G-Pon in telecom networks. It also produces epiwafers for PDs in telecom communication equipment.

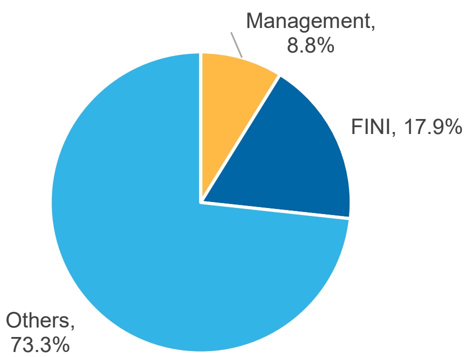
For the 'Other' segment, LandMark used to ship VCSELs for smartphone facial recognition to a leading US smartphone company. The solution is a GaAs-based laser, and due to the sheer size of the order, it consumed a substantial portion of LandMark's capacity. When the customer transitioned away from the VCSEL solution, LandMark suffered from underutilisation of its equipment, leading to loss-making years in 2023 and 2024.

Figure 11 - Landmark's product offerings



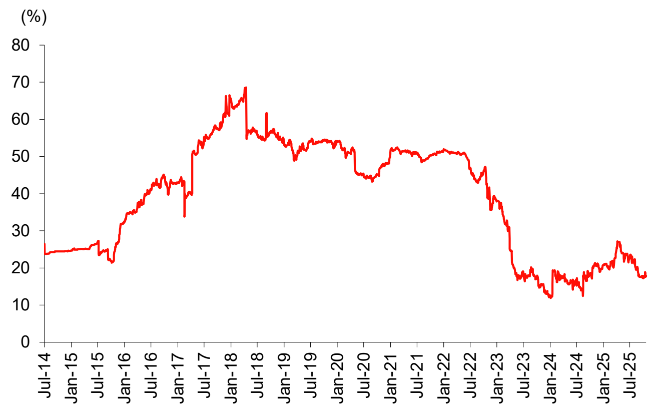
Source: Company, Nov 2025

Figure 12 - Shareholding structure



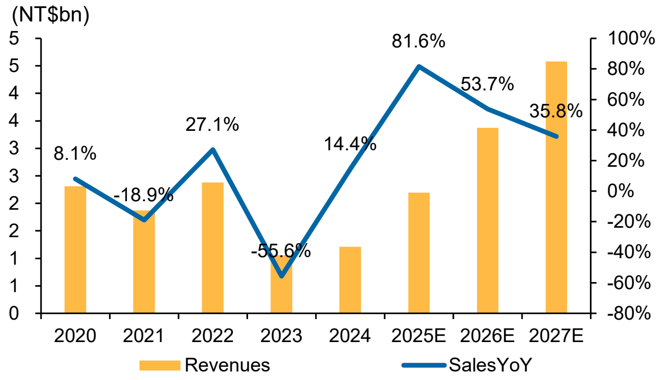
Source: TEJ, Macquarie Research, Nov 2025

Figure 13 - QFII holdings



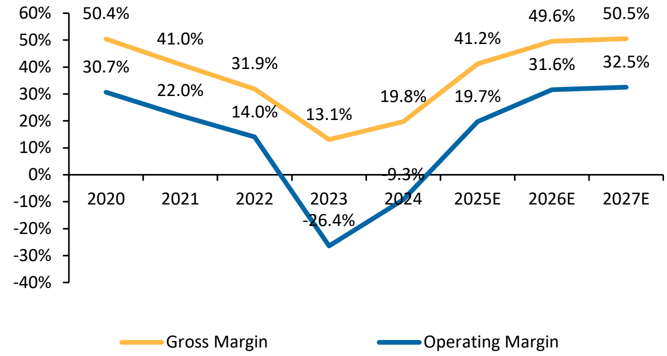
Source: TEJ, Macquarie Research, Nov 2025

Figure 14 - Sales / YoY growth



Source: Company, Macquarie Research, Nov 2025

Figure 15 - GM / OPM trends



Source: Company, Macquarie Research, Nov 2025

Figure 16 - P&L

NT$m	1Q25	2Q25	3Q25	4Q25E	1Q26E	2Q26E	3Q26E	4Q26E	2023	2024	2025E	2026E	2027E
Revenues	455	553	553	634	708	766	867	1,032	1,056	1,208	2,195	3,374	4,582
Gross Profit	147	240	241	276	344	376	430	522	138	239	903	1,673	2,315
Operating Profit	45	149	104	136	217	238	274	337	-279	-112	433	1,066	1,490
PBT	53	121	124	156	237	259	294	357	-264	-68	453	1,146	1,570
Net profit	42	97	99	125	190	207	235	285	-212	-55	363	917	1,256
EPS (NT$)	0.46	1.05	1.08	1.36	2.07	2.26	2.57	3.11	-2.31	-0.60	3.95	10.00	13.70
Profitability													
Gross Margin	32.3%	43.3%	43.6%	43.4%	48.6%	49.1%	49.6%	50.6%	13.1%	19.8%	41.2%	49.6%	50.5%
EBIT Margin	9.8%	26.9%	18.8%	21.4%	30.6%	31.1%	31.6%	32.6%	-26.4%	-9.3%	19.7%	31.6%	32.5%
Pretax Margin	11.6%	21.9%	22.4%	24.6%	33.5%	33.8%	33.9%	34.5%	-25.0%	-5.6%	20.7%	34.0%	34.3%
Net Margin	9.3%	17.5%	17.9%	19.6%	26.8%	27.0%	27.1%	27.6%	-20.0%	-4.5%	16.5%	27.2%	27.4%
QoQ Growth													
Sales	57.4%	21.7%	-0.1%	14.7%	11.7%	8.2%	13.2%	19.0%					
Operating Profit	-849.1%	232.9%	-30.4%	30.8%	60.1%	9.9%	14.9%	22.9%					
Pretax Profit	418.1%	129.0%	2.3%	25.8%	52.3%	9.0%	13.7%	21.3%					
Net Profit	418.1%	129.0%	2.3%	25.8%	52.3%	9.0%	13.7%	21.3%					
EPS	417.4%	129.0%	2.5%	25.8%	52.3%	9.0%	13.7%	21.3%					
YoY Growth													
Sales	40.3%	102.9%	71.1%	119.6%	55.8%	38.5%	56.9%	62.8%	-55.6%	14.4%	81.6%	53.7%	35.8%
Operating Profit	-167.2%	-414.5%	1161.8%	#######	384.7%	60.0%	164.3%	148.2%	-183.5%	-59.9%	-487.3%	146.1%	39.8%
Pretax Profit	na	na	1138.9%	1426.6%	348.7%	113.7%	137.6%	129.1%	na	na	na	152.9%	37.0%
Net Profit	na	na	1138.9%	1426.7%	348.7%	113.7%	137.6%	129.1%	na	na	na	152.9%	37.0%
EPS	na	na	1139.1%	1426.8%	349.4%	114.0%	137.6%	129.1%	na	na	na	153.0%	37.0%

Source: Company, Macquarie Research, Nov 2025

Figure 17 - Balance Sheet

NT$m	2019	2020	2021	2022	2023	2024	2025E	2026E	2027E
Cash & Cash Equivalents	1,341	1,547	1,771	1,644	581	529	339	655	1,659
Accounts Receivable	440	636	302	472	234	207	455	740	807
Inventories	193	208	289	376	357	346	625	889	974
Other Current Assets	11	17	18	20	736	1,043	1,314	1,314	1,314
TOTAL CURRENT ASSETS	**1,984**	**2,408**	**2,381**	**2,511**	**1,909**	**2,125**	**2,733**	**3,598**	**4,754**
Long-Term Investments	0	0	0	0	0	0	100	100	100
Gross Fixed Assets	3,396	4,453	4,671	5,007	5,316	5,418	5,711	6,071	6,431
Depreciation	(1,681)	(2,128)	(2,592)	(3,038)	(3,486)	(3,875)	(4,188)	(4,508)	(4,848)
Construction in Progress	759	61	44	195	138	122	0	0	0
Net Fixed Assets	2,474	2,387	2,123	2,164	1,969	1,665	1,522	1,562	1,582
Other L/T Assets	372	397	477	443	506	500	465	463	462
TOTAL ASSETS	**4,831**	**5,191**	**4,981**	**5,117**	**4,384**	**4,290**	**4,821**	**5,723**	**6,898**
Short-Term Debt	0	0	0	0	0	0	0	0	0
Accounts Payable	91	111	89	176	56	41	75	106	117
Current of Long-Term Debt	0	0	0	0	0	0	0	0	0
Other Current Liabilities	357	408	302	273	163	183	340	340	340
TOTAL CURRENT LIABILITIES	**448**	**519**	**391**	**450**	**219**	**225**	**415**	**447**	**457**
Long-Term Debt (ex. current portion)	0	0	0	0	0	0	0	0	0
Other Long Term Liabilities	316	323	318	311	303	294	289	289	289
TOTAL LIABILITIES	**764**	**842**	**709**	**761**	**522**	**519**	**704**	**736**	**746**
Common Stocks	910	914	914	919	919	925	925	925	925
Reserves	2,221	2,367	2,426	2,487	2,476	2,494	2,448	2,448	2,448
Retained Earnings	961	1,142	962	985	465	411	774	1,645	2,810
FX and Other Adjustments	(26)	(74)	(29)	(34)	2	(59)	(31)	(31)	(31)
Treasury Stock	0	0	0	0	0	0	0	0	0
Minority Interest	0	0	0	0	0	0	0	0	0
SHAREHOLDERS' EQUITY	**4,066**	**4,349**	**4,272**	**4,357**	**3,862**	**3,772**	**4,116**	**4,988**	**6,152**
TOTAL LIABILITIES AND EQUITY	**4,831**	**5,191**	**4,981**	**5,117**	**4,384**	**4,290**	**4,821**	**5,723**	**6,898**

Source: Company, Macquarie Research, Nov 2025

Figure 18 - Cash flow statement

NT$m	2019	2020	2021	2022	2023	2024	2025E	2026E	2027E
Net Income	450	591	338	330	(212)	(55)	363	917	1,256
Depreciation	388	457	486	473	455	389	313	320	340
Amortisation	5	6	6	4	3	2	2	2	2
Investment (Gain)/Losses - Equity	0	0	0	0	0	0	0	0	0
(Gain)/Losses from Disposals	0	0	0	0	0	(0)	0	0	0
Incr/(Decr) in Provisions	0	0	0	0	0	0	0	0	0
Net Change in Working Capital	159	(197)	231	(161)	127	27	(500)	(518)	(142)
Other	(102)	161	(39)	(6)	(133)	13	175	0	0
Cash from Operations	**901**	**1,017**	**1,022**	**639**	**240**	**377**	**352**	**721**	**1,456**
Capital Expenditures	(1,136)	(419)	(206)	(363)	(151)	(70)	(100)	(360)	(360)
Acquisition of L/T Investments	0	0	0	0	(719)	(300)	(351)	0	0
Proceeds from Disposal of S/T Investments	0	0	0	0	0	0	0	0	0
Other	(8)	(37)	(127)	(97)	(100)	(12)	(41)	0	0
Cash from Investing	**(1,144)**	**(456)**	**(333)**	**(460)**	**(970)**	**(382)**	**(492)**	**(360)**	**(360)**
Net Change in LT Debt	0	0	0	0	0	0	0	0	0
Net Change in ST Debt	0	0	0	0	0	0	0	0	0
Dividends Paid	(455)	(364)	(457)	(320)	(322)	(46)	(46)	(46)	(92)
Proceeds from Share Issuance	0	0	0	0	0	0	0	0	0
Employees & Director Bonus	0	0	0	0	0	0	0	0	0
Other Financing Cash Flows	(9)	11	(8)	14	(8)	(2)	(5)	0	0
Cash from Financing	**(463)**	**(353)**	**(465)**	**(306)**	**(330)**	**(48)**	**(51)**	**(46)**	**(92)**
Currency Adjustments	(1)	(2)	(0)	(1)	(3)	2	1	0	0
Net Cash Flow	**(708)**	**206**	**224**	**(128)**	**(1,063)**	**(52)**	**(190)**	**315**	**1,004**
Free Cash Flow	**(235)**	**598**	**816**	**276**	**89**	**306**	**252**	**361**	**1,096**

Source: Company, Macquarie Research, Nov 2025

Valuation

Our target price NT$480 is based on 35.0x 2027E EPS, similar to its 36.0x average forward multiple since 2022. We believe the valuation can be justified by the secular growth cycle during AI networking explosion and the shortage in optical chips. Our 2026/27E EPS is 9%/7% higher than Bloomberg consensus, as we believe investors underappreciate the explosive demand and the tight supply in upper stream components.

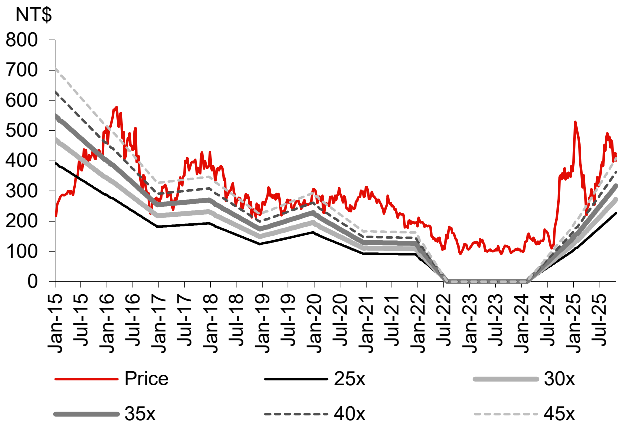
Taiwanese peers are currently trading at 26-49x forward P/E multiples (ignoring extreme values for FOCI and GCS given low earnings bases) due to 1) proximity to the world's largest AI hardware supply chain; 2) CPO's potential as a solution, which would circumvent geopolitical risk associated with China's dominance of the transceiver market, as CPO technology is less China-reliant and more dependent on Taiwan's semiconductor technology, and 3) rapid pace of revenue growth due to low base.

Figure 19 - Taiwanese optical communication names

			Mkt Cap	CP	TP	TSR	EPS (LC)			PER (X)			P/BV (X)			ROE (%)			Div yield (%)		
Ticker	Company	Rating	(US$m)	(LC)	(LC)	(%)	25E	26E	27E	25E	26E	27E	25E	26E	27E	25E	26E	27E	25E	26E	27E
Taiwan optical communication names																					
3081 TT	Landmark	O	1,222	410	480	17%	4.0	10.0	13.7	103.7	41.0	29.9	9.1	7.5	6.1	8.8	18.4	20.4	0.1	0.2	1.0
2455 TT	VPEC	O	888	149	236	64%	2.9	5.8	7.9	50.9	25.7	19.0	8.4	7.6	5.4	16.5	29.7	28.7	2.7	2.7	2.7
3363 TT	FOCI	NR	1,220	358	NA	NA	-0.5	0.8	2.7	n.a.	458.3	131.9	14.6	14.7	13.7	-2.1	3.2	10.8	0.0	0.2	0.3
4991 TT	GCS	NR	611	164	NA	NA	-2.1	0.2	5.4	n.a.	883.8	30.4	6.1	6.1	5.0	-7.9	-0.3	14.6	0.0	na	na
3450 TT	Elaser	NR	1,188	257	NA	NA	3.8	6.4	10.6	67.1	40.2	24.1	8.4	7.7	6.6	13.5	21.3	34.6	0.2	1.8	3.1
6442 TT	EZConn	NR	2,587	947	NA	NA	14.3	19.3	31.3	66.2	49.0	30.3	19.5	14.7	10.6	36.8	30.8	41.1	0.9	0.8	1.3

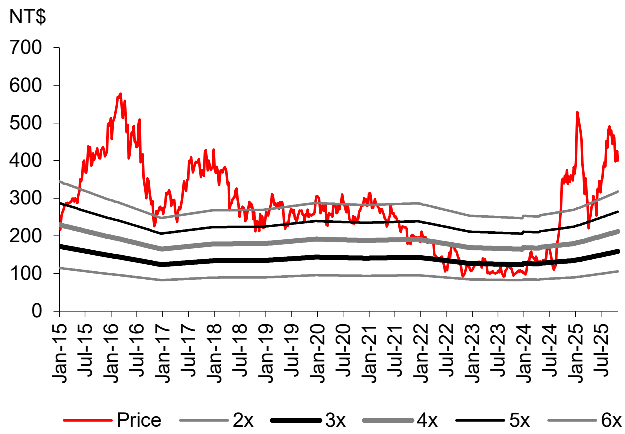
Source: Bloomberg, Macquarie Research, Nov 2025. Close prices as of 12 November

Figure 20 - 12-month forward P/E



Source: TEJ, Macquarie Research, Nov 2025

Figure 21 - 12-month forward P/B



Source: TEJ, Macquarie Research, Nov 2025

Risks to our investment thesis

- **Slower-than-expected transceiver demand**: A delay in Nvidia's next-gen GPU server might delay the adoption of AI networking equipment, thus potentially reducing our expected growth in demand for products such as CW lasers and the outsourcing PD market.
- **Excessive capacity at leading players:** Optical chips demand has been strong since 2024 and capacity expansion plans are on-going for leading players. Accelerated expansion plans across the industry could result in substantial oversupply and price erosion, a phenomenon with ample historical precedent.
- **Failed or delayed 3" qualification:** LandMark only has 2" capacity before the InP substrate ban, and CSP customers requires extensive testing (~6,000 hours for lasers before any product transition.) Failure or delays in securing qualification could lead to orders shift away from LandMark to other CW laser vendors, as customers continue to face pressure from demanding CSP and AI GPU customers, with their intense deployment roadmaps.

VPEC (2455 TT, OP, TP: NT$236 - PER)

- **Slower-than-expected transceiver demand**: A delay in Nvidia's next-gen GPU server might delay the adoption of AI networking equipments, with knock-on impacts on demand for optical chips and epiwafers.
- **Excessive capacity at leading players:** Optical chips demand has been strong since 2024 and capacity expansion plans are on-going for leading players. Accelerated expansion plans across the industry could result in substantial oversupply and price erosion, a phenomenon with ample historical precedent.
- **Declining wireless communication market:** A slower smartphone market will impact on VPEC's microelectronic business as it boasts 70% of market share in epiwafers for RF chips, leading to potential sales decline and margin erosion.

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Important Disclosures

Recommendation definitions	Volatility index definition	Financial definitions
Macquarie – Asia and USA Outperform – expected return >10% Neutral – expected return from -10% to +10% Underperform – expected return <-10% **Macquarie – Australia/New Zealand** Outperform – expected return >10% Neutral – expected return from 0% to 10% Underperform – expected return <0% During periods of share price volatility, recommendations and target prices may occasionally and temporarily be inconsistent with the above definitions. **Recommendations** – 12 months **12-month target** – Expected share price in 12 months **Valuation** – The company's estimated fair value share price based on the disclosed valuation methodology **Note**: Quant recommendations may differ from Fundamental Analyst recommendations	This is calculated from the volatility of historical price movements. **Very high** – highest risk – Stock should be expected to move up or down 60–100% in a year – investors should be aware this stock is highly speculative. **High** – stock should be expected to move up or down at least 40–60% in a year – investors should be aware this stock could be speculative. **Medium** – stock should be expected to move up or down at least 25–40% in a year. **Low** – stock should be expected to move up or down at least 15–25% in a year. * Applicable to select stocks in Asia/Australia/NZ **Note:** expected return is reflective of a Medium Volatility stock and should be assumed to adjust proportionately with volatility risk	All "Adjusted" data items have had the following adjustments made: **Added back:** goodwill amortisation, provision for catastrophe reserves, IFRS derivatives & hedging, IFRS impairments & IFRS interest expense Excluded: non recurring items, asset revals, property revals, appraisal value uplift, preference dividends & minority interests **EPS** = adjusted net profit / efpowa* **ROA** = adjusted ebit / average total assets **ROA Banks/Insurance** = adjusted net profit /average total assets **ROE** = adjusted net profit / average shareholders funds **Gross cashflow** = adjusted net profit + depreciation *equivalent fully paid ordinary weighted average number of shares All Reported numbers for Australian/NZ listed stocks are modelled under IFRS (International Financial Reporting Standards).

Recommendation proportions for quarter ending 30 Sept 2025

	AU/NZ	Asia	USA	
Outperform	51.96%	67.82%	67.16%	(for global coverage by Macquarie, 1.45% of stocks followed are investment banking clients)
Neutral	38.79%	18.36%	32.84%	(for global coverage by Macquarie, 1.75% of stocks followed are investment banking clients)
Underperform	9.25%	13.83%	0.00%	(for global coverage by Macquarie, 0.00% of stocks followed are investment banking clients)

Company-Specific Disclosures

Company Name	Disclosure
LandMark Optoelectronics (3081 TT) **Outperform** 12-month target: TWD480.00 - PER Valuation: TWD 480.00 - PER Price: TWD433.00	None
Visual Photonics Epitaxy (2455 TT) Valuation: TWD 236.00 - PER Price:TWD156.50	None

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